cisco

Lab Reference

MCP Quick Setup Guide

Connecting Claude, Cursor, and GitHub Copilot to Local and Web MCP Servers

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What is MCP?

The Model Context Protocol (MCP) lets an AI agent (Claude, Cursor, Copilot) connect to outside tools and data. There are two kinds of MCP servers:

- **Local server** — a program that runs on your own machine (usually via `npx`, `python`, or `docker`). The agent starts it for you. Good for file access, local databases, or anything that needs to touch your computer.
- **Web / remote server** — a hosted service you connect to by URL, often with a login (OAuth). No install required. Good for SaaS tools like GitHub, Notion, or Linear.

All three agents below support both kinds. The setup mechanics differ slightly per tool.

Before you start: most local servers run on Node.js — check you have it with `node --version` in a terminal (Terminal on Mac, Command Prompt on Windows). If missing, install the LTS version from nodejs.org.

Claude (Desktop app)

Local MCP server

1. Open Claude Desktop. Click the **Claude** menu in the menu bar (Mac) — *not* the in-app settings — and choose **Settings…**
2. Go to the **Developer** tab and click **Edit Config**. This creates (or opens) `claude_desktop_config.json`:
   - macOS: `~/Library/Application Support/Claude/claude_desktop_config.json`
   - Windows: `%APPDATA%\Claude\claude_desktop_config.json`
3. Add a server under `mcpServers`. Example — a filesystem server that can read/write your Desktop and Downloads folders:

```
{
  "mcpServers": {
    "filesystem": {
      "command": "npx",
      "args": [
        "-y",
        "@modelcontextprotocol/server-filesystem",
```

```
        "/Users/yourname/Desktop",
        "/Users/yourname/Downloads"
      ]
    }
  }
}
```

4. Save the file, then **fully quit and reopen** Claude Desktop.
5. Confirm it worked: click the **"Add files, connectors, and more"** icon (+) in the message box → **Connectors** → **Manage connectors**. You should see `filesystem` listed with its tools.

Web / remote MCP server

1. Click your profile name (bottom-left) → **Settings** → **Connectors**.
2. Click **Add custom connector** at the bottom of the list.
3. Paste the server's URL. Example — DeepWiki, a free public MCP server that answers questions about any public GitHub repo, no login required:

   `https://mcp.deepwiki.com/mcp`
4. If the server needs authentication, expand **Advanced settings** and enter the OAuth Client ID/Secret it gave you. Click **Add**, then **Connect** and complete the login popup if one appears.

> Free-tier Claude accounts are limited to one custom connector. Pro/Max/Team/Enterprise support more.

Cursor

Cursor uses one config format (`mcp.json`) for both local and remote servers — the only difference is whether you supply a `command` or a `url`.

File locations:

- Project-only: `.cursor/mcp.json` (inside the project folder)
- Global (all projects): `~/.cursor/mcp.json` (in your home folder)

You can also click **Settings → Tools & MCP → New MCP Server**, which opens this same file for you.

Local MCP server

```
{
  "mcpServers": {
    "filesystem": {
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-filesystem", "/Users/yourname/Desktop"]
    }
  }
}
```

Web / remote MCP server

Use `url` instead of `command`/`args`. Example — the same DeepWiki server, which needs no headers or auth:

```
{
  "mcpServers": {
    "deepwiki": {
```

```
      "url": "https://mcp.deepwiki.com/mcp"
    }
  }
}
```

For a server that requires a token, add a `headers` object:

```
{
  "mcpServers": {
    "github": {
      "url": "https://api.githubcopilot.com/mcp/",
      "headers": { "Authorization": "Bearer YOUR_TOKEN" }
    }
  }
}
```

After saving, open **Customize** in the sidebar to confirm the server is enabled (green) and trusted. Cursor will prompt for an OAuth login in your browser the first time you use a server that requires it.

GitHub Copilot (VS Code)

Copilot's MCP support lives inside VS Code. **Agent mode must be on**: in Copilot Chat, use the mode dropdown and select **Agent**.

File locations (note the root key is `servers`, not `mcpServers`):

- Project-only: `.vscode/mcp.json`
- Global (all workspaces): run **MCP: Open User Configuration** from the Command Palette (Cmd+Shift+P on Mac, Ctrl+Shift+P on Windows/Linux)

You can also run **MCP: Add Server** from the Command Palette for a guided setup, or install ready-made servers from the Extensions view by searching `@mcp`.

Local MCP server

```
{
  "servers": {
    "filesystem": {
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-filesystem", "/Users/yourname/Desktop"]
    }
  }
}
```

Web / remote MCP server

Add `"type": "http"` and a `url`. Example combining both kinds in one file:

```
{
  "servers": {
    "deepwiki": {
      "type": "http",
      "url": "https://mcp.deepwiki.com/mcp"
    },
    "playwright": {
      "command": "npx",
```

```
      "args": ["-y", "@microsoft/mcp-server-playwright"]
    }
  }
}
```

The first time a server starts, VS Code shows a **trust prompt** — review the config and confirm before it runs. Avoid hardcoding API keys directly in `mcp.json`; use `${input:...}` variables instead so you're prompted securely.

Cheat sheet

Tool	Local setup	Remote setup	Root key
Claude Desktop	Settings → Developer → Edit Config	Settings → Connectors → Add custom connector	`mcpServers`
Cursor	Edit `.cursor/mcp.json`, use `command/args`	Same file, use `url`	`mcpServers`
Copilot (VS Code)	Edit `.vscode/mcp.json`, use `command/args`	Same file, add `type: "http"` + `url`	`servers`

Security tips

Only add servers from sources you trust — a local server can run arbitrary code on your machine, and a remote server can act on your accounts. Keep API keys and tokens in environment variables or your OS keychain rather than pasted directly into config files, and review what data or scopes a server requests before connecting.

Troubleshooting

- **Claude Desktop:** logs are at `~/Library/Logs/Claude` (Mac) or `%APPDATA%\Claude\logs` (Windows). Always fully quit/reopen the app after editing the config.
- **Cursor:** Output panel (Cmd/Ctrl+Shift+U) → select **MCP Logs**.
- **Copilot (VS Code):** Command Palette → **MCP: List Servers** → select a server → **Show Output**.
- All three: if a server doesn't appear, double-check the JSON is valid (no trailing commas) and that you used the right root key (`mcpServers` vs `servers`).